

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences and CGISS, Boise State University
Environmental Research Building (ERB) 1104 (lab), 3155 (office)
1910 University Drive
Boise, ID 83725

(208) 426-1416 (office)
(303) 859-3106 (cell)
hpmarshall@boisestate.edu
cgiss.boisestate.edu/~hpm

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Selected Professional Experience

- 08/2013-present Associate Professor, Department of Geosciences and Center for Geophysical Investigation of the Shallow Subsurface (CGISS), Boise State University
- 06/2018-present Project Scientist, NASA SnowEx 2020 Mission
- 07/2007-present Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire

Honors and Awards

- 2019 2018 WRR Editors' Choice Award: Hedrick et al., 2018 (recent PhD student)
- 2018 NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
- 2012 Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
- 2010-13 New Investigator Program Award, NASA, 1st in Idaho history
- 2010 Young Scientist Award, Cryosphere Section, American Geophysical Union
- 2009 Top Reviewer Award, Cold Regions Science and Technology (CRST)
- 2008 U.S. Young Scientist Award, International Union of Radio Science (URSI)
- 2002-05 N.A.S.A. Earth Systems Science Graduate Fellowship
- 2001-02 N.S.F. K-12 Engineering Outreach Fellowship
- 1999 Volunteer of the Year, Arctic Canada, Frontiers Foundation

Student Awards

- 2019 Joel Gongora, Ph.D. student, *National NASA Earth and Space Sciences Fellowship*
- 2018 Tate Meehan, Ph.D. student, *Idaho Space Grant Consortium Fellowship*
- 2018 Tate Meehan, Ph.D. student *U.S. Army CRREL Student Career Experiences Program*
- 2012 Andrew Hedrick, M.S. student, *American Avalanche Association Graduate Student Grant*
- 2012 Scott Havens, Ph.D. student, 1st place poster *NSF/NASA EPSCoR annual meeting*
- 2011 Scott Havens, Ph.D. Student, *National NASA Earth and Space Sciences Fellowship*
- 2012 Kris Burch, Lloyd Lowe II, Kolton Drake, *Best Senior Project, BSU College of Engineering*
- 2003 Adam Maerz, High School student, *Colorado State Science Fair Finalist*

Selected Refereed Student and Post-doc Authored Publications

[h-index=23, i10-index=40, citations=1525]

Primary advisor: 1 Post-doc, 5 PhD, 7 MS, 13 undergrad; served on 13 PhD and 15 MS committees

Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and **H.P. Marshall** (in review). Within-stand boundary effects of snow water equivalent distribution in forested areas. *Water Resources Research*

Robertson, M., **H.P. Marshall**, Roberts, D., and Kunkel, M. (in review). Continuous snow water equivalent measurements with ultra-wideband impulse radar: A new sensor and time-delay method of processing downward-looking radar. *Hydrological Processes*

Hedrick, A. R., Marks, D., Marshall, H.-P., McNamara, J., Havens, S., Trujillo, E., Sandusky, M., Robertson, M., Johnson, M., Bormann, K. J., and Painter, T. H. (in review). From drought to flood: A water balance analysis of the tuolumne river basin during extreme conditions (2015 – 2017). *Hydrol. Process.*

Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12

Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. Z. U. R. (2019). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318

Meehan, T. G., Marshall, H.-P., Deeb, E. J., and Shoop, S. A. (2019). Snowmicropenetrometer applications for winter vehicle mobility. ERDC 6.2 Boreal Aspects of Ensured Maneuver (BAEM) TR-19-14, U.S. Army Cold Regions Research and Engineering Laboratory

Lewis, G., Osterberg, E., Hawley, R., Marshall, H. P., Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., and Ferris, D. (2019). Recent precipitation decrease across the western greenland ice sheet percolation zone. *The Cryosphere*, 13:2797–2815

Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., **H.P. Marshall**, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. *Geophysical Research Letters*, 45(7):3164–3172

Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063

Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **H.P. Marshall**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Geoscience and Remote Sensing*, 66(5):2480–2494

Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., **H.P. Marshall**, and Box, J. (2017). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. *The Cryosphere*, 11(2):773

Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154

Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248

Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23

- Havens, S., **Marshall, H.P.**, Trisca, G., and Johnson, J. B. (2014c). Real time avalanche detection for high risk areas. Monograph FHWA-ID-14-219, Idaho Transportation Department, 3311 W State Street, P. O. Box 7129, Boise, ID 83707-1129 United States
- Kormos, P. R., Marks, D., McNamara, J. P., **Marshall, H.P.**, Winstral, A., and Flores, A. N. (2014a). Snow distribution, melt and surface water inputs to the soil in the mountain rain-snow transition zone. *Journal of Hydrology*, 519:190–204
- Walters, R. D., Watson, K. A., **Marshall, H.P.**, McNamara, J. P., and Flores, A. N. (2014). A physiographic approach to downscaling fractional snow cover data in mountainous regions. *Remote Sensing of Environment*, 152:413–425
- Havens, S., **Marshall, H.P.**, Johnson, J. B., and Nicholson, B. (2014b). Calculating the velocity of a fast-moving snow avalanche using an infrasound array. *Geophysical Research Letters*, 41(17):6191–6198
- Anderson, B. T., McNamara, J. P., **Marshall, H.P.**, and Flores, A. N. (2014). Insights into the physical processes controlling correlations between snow distribution and terrain properties. *Water Resources Research*, 50(6):4545–4563
- Tinkham, W. T., Smith, A. M. S., **Marshall, H.P.**, Link, T. E., Falkowski, M. J., and Winstral, A. H. (2014). Quantifying spatial distribution of snow depth errors from lidar using random forest. *Remote Sensing of Environment*, 141:105–115
- McCreight, J. L., Slater, A. G., **Marshall, H.P.**, and Rajagopalan, B. (2014). Inference and uncertainty of snow depth spatial distribution at the kilometre scale in the colorado rocky mountains: the effects of sample size, random sampling, predictor quality, and validation procedures. *Hydrological Processes*, 28(3):933–957
- Kormos, P. R., Marks, D., Williams, C. J., **Marshall, H.P.**, Aishlin, P., Chandler, D. G., and McNamara, J. P. (2014b). Soil, snow, weather, and sub-surface storage data from a mountain catchment in the rain-snow transition zone. *Earth System Science Data*, 6(1):165–173
- Havens, S., **Marshall, H.P.**, Pielmeier, C., and Elder, K. (2013b). Automatic grain type classification of snow micro penetrometer signals with random forests. *IEEE Transactions On Geoscience and Remote Sensing*, 51(6):3328–3335
- Eiriksson, D., Whitson, M., Luce, C. H., **Marshall, H.P.**, Bradford, J., Benner, S. G., Black, T., Hetrick, H., and McNamara, J. P. (2013). An evaluation of the hydrologic relevance of lateral flow in snow at hillslope and catchment scales. *Hydrological Processes*, 27(5):640–654
- Tinkham, W. T., Hoffman, C. M., Falkowski, M. J., Smith, A., **Marshall, H.P.**, and Link, T. E. (2013a). A methodology to characterize vertical accuracies in lidar-derived products at landscape scales. *Photogrammetric Engineering & Remote Sensing*, 79(8):709–716
- Mikesell, T. D., van Wijk, K., Haney, M. M., Bradford, J. H., **Marshall, H.P.**, and Harper, J. T. (2012). Monitoring glacier surface seismicity in time and space using rayleigh waves. *Journal of Geophysical Research-Earth Surface*, 117(F2):F02020
- Geroy, I., Gribb, M., **Marshall, H.P.**, Chandler, D., Benner, S. G., and McNamara, J. P. (2011). Aspect influences on soil water retention and storage. *Hydrological Processes*, 25(25):3836–3842
- Tape, K. D., Rutter, N., **Marshall, H.P.**, Essery, R., and Sturm, M. (2010b). Recording microscale variations in snowpack layering using near-infrared photography. *Journal of Glaciology*, 56(195):75–80
- Tape, K. D., Lord, R., **Marshall, H.P.**, and Ruess, R. W. (2010a). Snow-mediated ptarmigan browsing and shrub expansion in arctic alaska. *Ecoscience*, 17(2):186–193
- Lutz, E., Birkeland, K., and **Marshall, H.P.** (2009b). Quantifying changes in weak layer microstructure associated with artificial load changes. *Cold Regions Science and Technology*, 59(2-3):202–209